RAMAIAH
Institute of Technology

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSP67: MINI PROJECT **TERM: FEB - MAY 2026**

PROJECT SYNOPSIS

17/02/2026

Submitted to
Dr. Dayananda R. B.

PROJECT TEAM MEMBERS

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M.S. RAMAIAH INSTITUTE OF TECHNOLOGY
(Autonomous Institute, Affiliated to VTU)

Title of the Project:

HoloScript — AI-Driven Real-Time 3D Hologram Simulation with Voice and Gesture Control

Project Stream:

Visual Intelligence / Data Intelligence / Multimedia Systems

Problem Statement:

Most existing student-level hologram or 3D visualization systems rely on pre-designed, static models and prerecorded animations. They do not support real-time content generation from natural language input, nor do they allow intuitive interaction without traditional input devices. This limits accessibility, adaptability, and personalization in educational and visualization systems.

There is a need for an intelligent, real-time 3D visualization platform that can dynamically generate scenes from user voice commands and allow natural gesture-based interaction — without requiring specialized hardware — so that complex spatial concepts can be explored interactively and intuitively.

Objective:

1. To design and develop an AI-assisted software system that converts voice commands into structured 3D scene descriptions using a large language model.
2. To build a real-time 3D rendering pipeline that generates and animates parametric scenes dynamically instead of using prebuilt models.
3. To implement webcam-based hand gesture recognition for touch-free interaction such as rotation, scaling, and scene transformation.
4. To create a simulated hologram display output and frame conversion pipeline compatible with a future physical volumetric display (major project phase).
5. To integrate live data sources (such as weather or scientific datasets) into dynamic 3D visualizations.

Scope of the Project:

This mini project focuses entirely on the software intelligence and visualization pipeline. The system will run on a standard laptop without external hardware. It includes voice input processing, AI-based scene generation, 3D rendering, gesture recognition, and simulated hologram visualization through a desktop dashboard.

The project scope includes real-time rendering, gesture control, and AI-generated parametric scene construction. Physical holographic hardware implementation is planned as the major project extension. The software architecture will be designed so that its output frame format can directly drive a hardware volumetric display in the next phase.

Hardware & Software:

Hardware:

- Standard Laptop/Desktop
- Webcam (for gesture tracking)
- Microphone (for voice input)

Software:

- Python 3.x
- OpenAI Whisper (offline speech-to-text)
- LLM API (for structured scene generation)
- MediaPipe Hands (gesture recognition)
- PyOpenGL + Pygame (3D rendering)
- FastAPI + WebSockets (real-time streaming backend)
- SQLite (local scene history storage)
- REST APIs for live data integration

What contribution to the society would the project make?

This project enables intuitive, interactive 3D visualization using voice and gesture input, which can significantly improve how complex spatial concepts are taught and understood. It supports educational visualization in subjects such as chemistry, astronomy, and engineering and medical fields by allowing users to generate and manipulate 3D models naturally instead of relying on static diagrams.

The system also promotes touch-free interaction, which would be useful in smart classrooms, museums, and public information systems. By designing a low-cost, software-first architecture that can later connect to affordable holographic hardware, the project contributes toward democratizing advanced visualization technology for educational and research use.

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Guide Comments:

Signature of the Guide with date

Expert Committee Reviews

Sl. No		**Comments**
1	Expert 1	
2	Expert 2	

Accepted ☐ **Rejected** ☐